Nonlinear Manifold Approximation for Constraint Modes in Substructuring

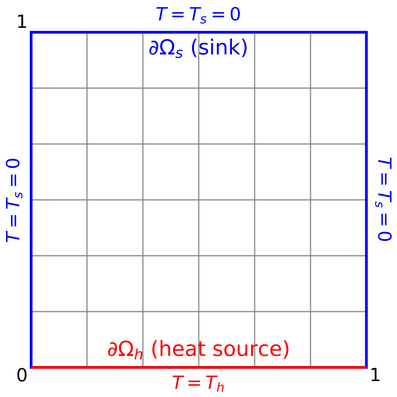
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Introduction

The nonlinear heat equation is solved on a rectangular domain with smooth random boundary inputs. The diffusivity law is prescribed as $\kappa(T;\alpha) = T^\alpha$, where α controls the nonlinearity.

$$\begin{cases} \nabla \cdot (\kappa(T;\alpha)\nabla T) = 0 & \mathbf{x} \in \Omega \\ T = T_h & \mathbf{x} \in \partial\Omega_h \\ T = T_s & \mathbf{x} \in \partial\Omega_s \end{cases} \tag{1}$$



The constraint mode $\mathcal{C}$ is traditionally defined for linear structures. However, it may not be adequate for nonlinear structures. Therefore, the goal is to learn a nonlinear correction term using either a quadratic or a neural network model.

$$u \approx \mathcal{C}q + \mathcal{H}q \otimes q \quad \text{or} \quad u \approx \mathcal{C}q + \mathcal{N}(q)$$

The inputs will be T_h, and the outputs are the solution field.

Data Collection

- Equation (1) is solved using a second-order finite difference scheme.
- Ten different $\alpha \in [0, 2]$ are selected based on log spacing.
- T_h is generated based on a Fourier series $T_h = \sum_{k=1}^{5} a_k \sin^2(k\pi s)$ — where a_k are random coefficients — and normalized to be between $[0, 1]$.
- For each α, 120 snapshots corresponding to different T_h are generated. 100 are reserved for training, and 20 for testing.

Background: Substructuring

In structural mechanics, substructuring is often employed to perform domain decomposition and speed up finite element computations. To couple the substructures, the constraint mode

$$\mathcal{C} = \begin{bmatrix} -K_{ii}^{-1} K_{i\Gamma} \\ I \end{bmatrix}$$

ensures continuity between substructure interfaces with I on the interface block row.

- The nonlinear heat equation (1) mimics the structural mechanics PDE. (Both elliptic)
- $\mathcal{C}$ is computed for the nonlinear heat equation by applying unit excitations to $\partial\Omega_h$.

Dimensionality Reduction of T_h

- The reduced coordinate $q \in \mathbb{R}^r$ is computed for each boundary input T_h.
- Define the collection of all n boundary inputs $B = [T_{h,1}, \ldots, T_{h,n}] \in \mathbb{R}^{N_h \times n}$.
- The singular value decomposition of B gives a low rank approximation $B \approx U_{(1:r)}\Sigma_{(1:r)}V_{(1:r)}^\top$ where $r \ll n$. Then we define $q = V^\top T_h$ with $V = U_{(1:r)}$.
- Model feature is reduced from $\mathcal{O}(n^2)$ to $\mathcal{O}(r^2)$ for the quadratic model.

Method 1: Quadratic Operator Inference

The quadratic model is $u \approx \mathcal{C}q + \mathcal{H}q \otimes q$, where $\mathcal{H}$ is learned from the residual $u - \mathcal{C}q$:

$$\mathcal{H} = \arg\min_{\mathcal{H}} \sum_k \|\mathcal{H}q_k \otimes q_k - (u - \mathcal{C}q_k)\|_2^2 + \lambda\|\mathcal{H}\|_2^2 \tag{2}$$

A regularization term is added to prevent overfitting.

Method 2: Neural Network Model

Instead of a quadratic model, one can swap it with a neural network term $\mathcal{N}(q)$ instead:

$$u \approx \mathcal{C}q + \mathcal{N}(q) \tag{3}$$

where $\mathcal{N}$ is a mapping from q to the residual terms $u - \mathcal{C}q$. This is implemented with `sklearn` as a simple two-layer MLP with 64 neurons in each layer. Moreover, the activation function for each neuron is chosen as $\tanh$ because it is differentiable, which is desirable since the underlying physics is differentiable.

Solution Field Comparison at $\alpha = 1.031$

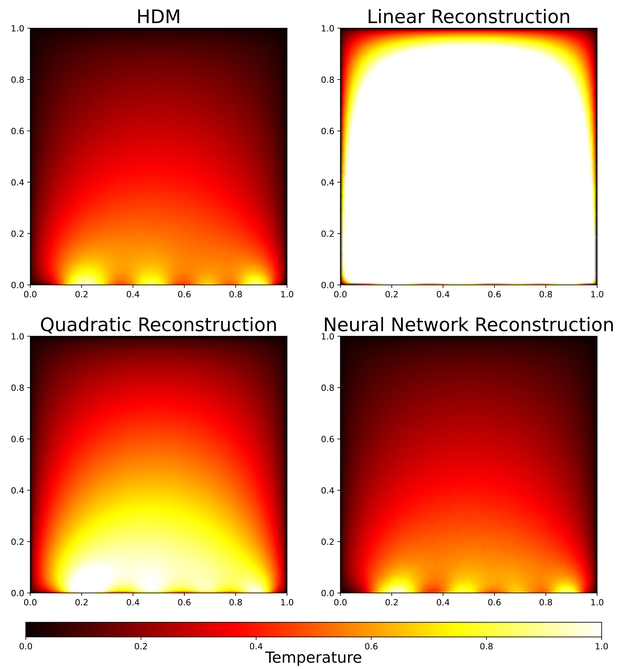


Figure 1. Solution Comparison for $\alpha = 1.031$

Numerical Results

The training workflow can be summarized:

Data Compression: $B \xrightarrow{\text{SVD}} V$ and $q_i = V^\top T_{h,i}$

Model Training: $\{q_i, u_i\}_{i=1}^n \longrightarrow$ Model Training $\longrightarrow$ obtain H or $\mathcal{N}(q)$

Online approximation: some new $\bar{T}_h \xrightarrow{V^\top} \bar{q} \longrightarrow$ Model Evaluation (Quadratic or Neural-Net) $\longrightarrow u$

The solution comparison at $\alpha = 1.031$ is shown in Fig. 1:

- Linear reconstruction: completely misrepresents HDM.
- Quadratic reconstruction: captures overall shape, misses details.
- Neural network: nearly perfect, only misses fine features near $\partial\Omega_h$.

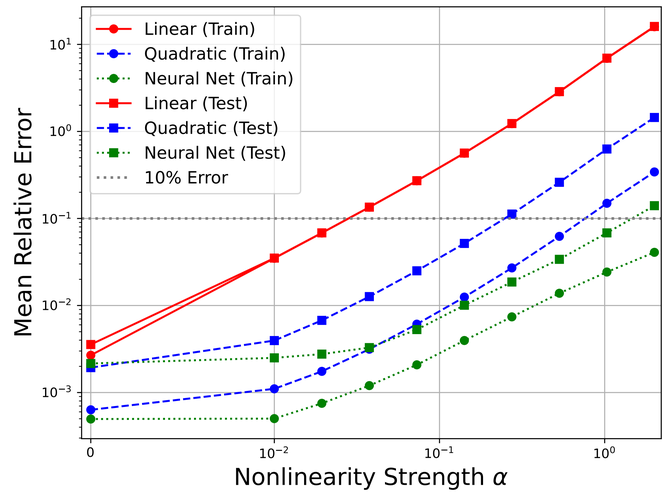


Figure 2. Mean Relative Error of the 20 Test T_h across all α and all models

Discussion

- Shown in Fig. 2, the neural network outperforms both linear and quadratic models.
- This is expected because the solution manifold of the PDE (1) is not necessarily linear or quadratic; thus, a more flexible neural network would stand out.
- Interestingly, the error growth rate is very similar between the three models.

In the future, I would like to extend this into a fully intrusive reduced-order model for the nonlinear heat equation.

References

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